

Milestone 4: Infectious Disease Outbreak in California

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Introduction

Scenario 1: Infectious Disease Outbreak (Simulated) in California

The `sim_novelid_CA.csv` file and `sim_novelid_LACounty.csv` file together represent simulated morbidity for the entire state of California. This analysis combines and analyzes these datasets to understand the temporal and demographic patterns of the outbreak.

Objectives

This milestone focuses on:

- Final datasets for creation of visualization
- Analyzing outbreak patterns over time by demographic strata
- Identifying populations at highest risk

Data Preparation

Load and Clean Morbidity Data

```
# Load datasets
ca_pop <- read_csv(here("outbreak_data", "ca_pop_2023.csv"))
sim_novelid_ca <- read_csv(here("outbreak_data", "sim_novelid_CA.csv"))
sim_novelid_la <- read_csv(here("outbreak_data", "sim_novelid_LACounty.csv"))

# Clean column names to snakecase
sim_novelid_ca <- sim_novelid_ca %>% clean_names()
sim_novelid_la <- sim_novelid_la %>% clean_names()
```

Standardize Column Names and Select Variables

```
# Standardize CA dataset
simca1 <- sim_novelid_ca %>%
  select(
    county,
    age_cat,
    sex,
    race_eth = race_ethnicity,
    dt_diagnosis,
    time_int,
    new_infections,
    cumulative_infected,
```

```

    new_severe,
    cumulative_severe
  )

# Standardize LA County dataset
simla1 <- sim_novelid_la %>%
  mutate(county = "Los Angeles") %>%
  select(
    county,
    age_cat = age_category,
    sex,
    race_eth,
    dt_diagnosis = dt_dx,
    new_infections = dx_new,
    cumulative_infected = infected_cumulative,
    new_severe = severe_new,
    cumulative_severe = severe_cumulative
  )

```

Handle Missing Data

```

# Remove rows with missing values to ensure clean joins
simca1 <- simca1 %>% drop_na()
simla1 <- simla1 %>% drop_na()

```

Dates and CDC Week

CDC epidemiological weeks start on Sunday and are the standard for disease surveillance reporting.

```

# Fix date format for LA County
simla1 <- simla1 %>%
  mutate(dt_diagnosis = dmy(dt_diagnosis))

# CDC has a Sunday start date
set_week_start(7)

# Create CDC week for CA dataset from time_int variable
simca2 <- simca1 %>%
  mutate(
    year = substr(time_int, 1, 4),
    week = substr(time_int, 5, 6),
    cdc_week = aweek::get_date(week = week, year = year)
  ) %>%
  select(-year, -week)

# Create CDC week for LA County from diagnosis date
simla2 <- simla1 %>%
  mutate(
    cdc_week = get_date(
      week = isoweek(dt_diagnosis),
      year = year(dt_diagnosis)
    )
  )

```

Recode Race/Ethnicity Categories

```
# Convert numeric race codes to descriptive labels for CA data
simca3 <- simca2 %>%
  mutate(
    race_eth = recode_factor(race_eth,
                             `1` = "White, Non-Hispanic",
                             `2` = "Black, Non-Hispanic",
                             `3` = "American Indian or Alaska Native, Non-Hispanic",
                             `4` = "Asian, Non-Hispanic",
                             `5` = "Native Hawaiian or Pacific Islander, Non-Hispanic",
                             `6` = "Multiracial (two or more of above races), Non-Hispanic",
                             `7` = "Hispanic (any race)",
                             `9` = "Unknown"
    )
  ) %>%
  select(-time_int)

# Ensure race is factor type for LA data
simla3 <- simla2 %>%
  mutate(race_eth = factor(race_eth))
```

Combine Datasets

```
# Merge CA and LA County morbidity data
sim_all <- bind_rows(simca3, simla3)
```

Population Data Preparation

Clean and Standardize Population Data

```
# Select and rename columns to match morbidity data
ca_pop1 <- ca_pop %>%
  select(
    county,
    age_cat,
    sex,
    race_eth = race7,
    pop
  )

# Standardize race/ethnicity labels to match morbidity data
ca_pop2 <- ca_pop1 %>%
  mutate(
    race_eth = recode_factor(race_eth,
                             "WhiteTE NH" = "White, Non-Hispanic",
                             "Black NH" = "Black, Non-Hispanic",
                             "AIAN NH" = "American Indian or Alaska Native, Non-Hispanic",
                             "Asian NH" = "Asian, Non-Hispanic",
                             "NHPI NH" = "Native Hawaiian or Pacific Islander, Non-Hispanic",
                             "MR NH" = "Multiracial (two or more of above races), Non-Hispanic",
                             "Hispanic" = "Hispanic (any race)"
    )
  )
```

```

)

# Reclassify age categories to match morbidity data (e.g., 0-17 instead of 0-4, 5-11, 12-17)
ca_pop3 <- ca_pop2 %>%
  mutate(age_cat = case_when(
    age_cat %in% c("0-4", "5-11", "12-17") ~ "0-17",
    TRUE ~ age_cat
  )) %>%
  group_by(county, age_cat, sex, race_eth) %>%
  summarise(pop = sum(pop, na.rm = TRUE)) %>%
  ungroup()

```

Standardize County Names

```

# Remove "County" suffix from morbidity data to match population data
sim_all1 <- sim_all %>%
  mutate(
    county = str_to_title(str_remove(county, " County$")),
    county = str_trim(county)
  )

# Ensure consistent formatting in population data
ca_pop4 <- ca_pop3 %>%
  mutate(
    county = str_to_title(str_trim(county))
  )

```

Join Morbidity and Population Data

```

# Left join to add population data to morbidity data
sim_pop <- sim_all1 %>%
  left_join(ca_pop4,
    by = c("county", "age_cat", "sex", "race_eth"))

```

Time Series Analysis

Calculate Weekly Rates

```

# Create dataset with weekly cumulative incidence rates and drop columns
fun1 <- sim_pop %>%
  mutate(
    ci_per100k = (new_infections / pop) * 100000,
    severe_per100k = (new_severe / pop) * 100000
  ) %>%
  select(-c(dt_diagnosis, cumulative_infected, cumulative_severe))

```

Overall California Outbreak Trajectory

```

# Aggregate to state level by week
ca_overall <- fun1 %>%
  group_by(cdc_week) %>%

```

```

summarise(
  total_cases = sum(new_infections, na.rm = TRUE),
  total_pop = sum(pop, na.rm = TRUE),
  total_severe = sum(new_severe, na.rm = TRUE)
) %>%
mutate(
  ci_per100k = (total_cases / total_pop) * 100000
) %>%
ungroup()

```

Stratify by Age Group

```

# Aggregate by age group and week
ca_by_age <- fun1 %>%
  group_by(cdc_week, age_cat) %>%
  summarise(
    total_cases = sum(new_infections, na.rm = TRUE),
    total_pop = sum(pop, na.rm = TRUE),
    total_severe = sum(new_severe, na.rm = TRUE)
  ) %>%
  mutate(
    ci_per100k = (total_cases / total_pop) * 100000
  ) %>%
  ungroup()

```

Stratify by Race/Ethnicity

```

# Aggregate by race/ethnicity and week
ca_by_race <- fun1 %>%
  group_by(cdc_week, race_eth) %>%
  summarise(
    total_cases = sum(new_infections, na.rm = TRUE),
    total_pop = sum(pop, na.rm = TRUE),
    total_severe = sum(new_severe, na.rm = TRUE)
  ) %>%
  mutate(
    ci_per100k = (total_cases / total_pop) * 100000
  ) %>%
  ungroup()

```

Results

Epidemic Curve by Age Group

The outbreak affected all age groups, with clear differences in disease burden by age.

```

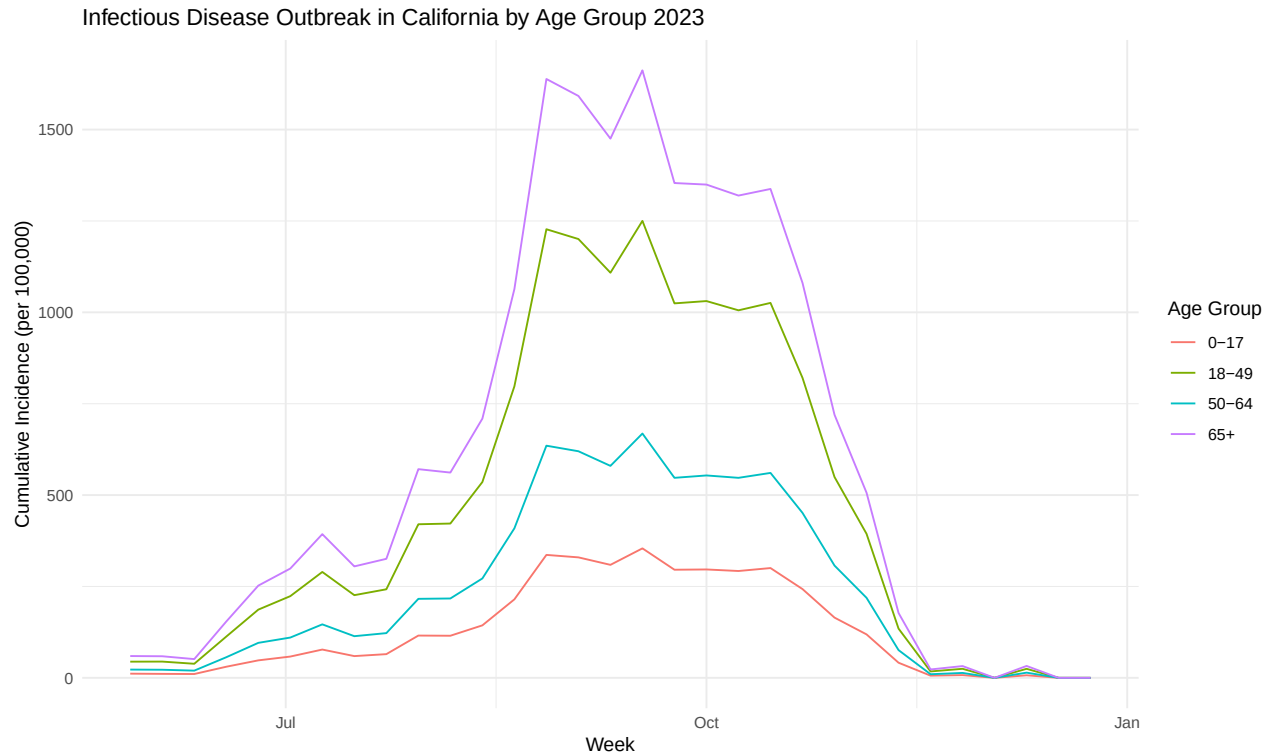
ggplot(ca_by_age, aes(x = cdc_week,
                      y = ci_per100k,
                      color = age_cat)) +
  geom_line() +
  labs(
    title = "Infectious Disease Outbreak in California by Age Group 2023",

```

```

x = "Week",
y = "Cumulative Incidence (per 100,000)",
color = "Age Group"
) +
theme_minimal()

```



Key Findings:

- All age groups peaked simultaneously around August-September 2023.
- Incidence increases with age, however, cumulative incidence for 18-49 > 50-64.
- Persons aged 65+ experienced the highest burden (~1,600 per 100,000 at peak).
- Persons aged 0-17 had the lowest incidence (~350 per 100,000 at peak).

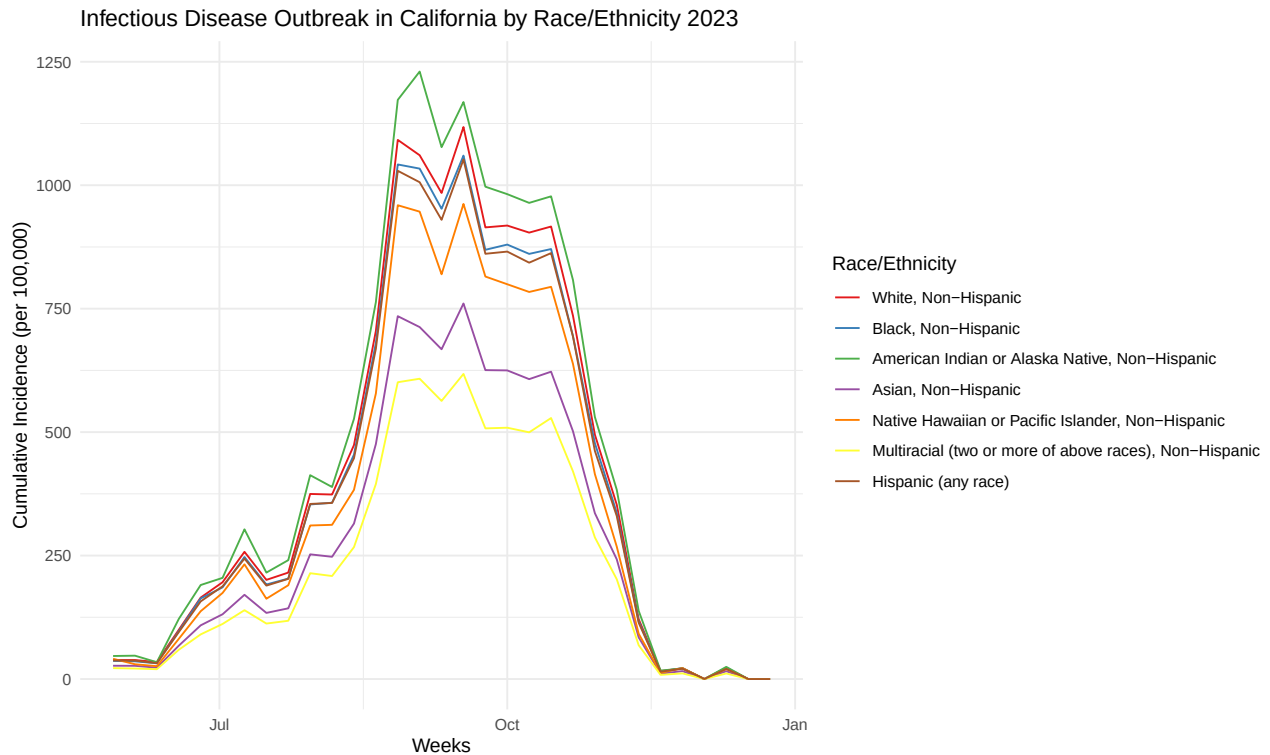
Epidemic Curve by Race/Ethnicity

```

ggplot(ca_by_race, aes(x = cdc_week,
                        y = ci_per100k,
                        color = race_eth)) +

geom_line() +
scale_color_brewer(palette = "Set1") +
labs(
  title = "Infectious Disease Outbreak in California by Race/Ethnicity 2023",
  x = "Weeks",
  y = "Cumulative Incidence (per 100,000)",
  color = "Race/Ethnicity"
) +
theme_minimal()

```



Key Findings:

- American Indian or Alaska Native populations experienced the highest peak incidence (~1,250 per 100,000).
- All racial/ethnic groups followed similar temporal patterns, peaking around the same time.

Summary Statistics

Summary by Age Group

```
summary_age <- ca_by_age %>%
  group_by(age_cat) %>%
  summarise(
    total_cases = sum(total_cases, na.rm = TRUE),
    avg_population = mean(total_pop, na.rm = TRUE), # Total population is the same every week
    peak_weekly_ci = max(ci_per100k, na.rm = TRUE),
    total_severe = sum(total_severe, na.rm = TRUE)
  ) %>%
  mutate(
    attack_rate_percent = (total_cases / avg_population) * 100,
    severe_rate_percent = (total_severe / total_cases) * 100
  )

kable(summary_age,
  digits = 2,
  col.names = c("Age Group", "Total Cases", "Average Population",
    "Peak Weekly CI (per 100k)", "Total Severe Cases",
    "Attack Rate (%)", "Severe Rate (%)"),
  caption = "Outbreak Summary Statistics by Age Group") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
```

```
full_width = FALSE)
```

Table 1: Outbreak Summary Statistics by Age Group

Age Group	Total Cases	Average Population	Peak Weekly CI (per 100k)	Total Severe Cases	Attack Rate (%)
0-17	353210	8694111	354.22	363	4.06
18-49	2467284	17106505	1250.22	29325	14.42
50-64	540481	7086797	668.40	2822	7.63
65+	1188608	6221657	1662.00	94416	19.10

Summary by Race/Ethnicity

```
summary_race <- ca_by_race %>%
  group_by(race_eth) %>%
  summarise(
    total_cases = sum(total_cases, na.rm = TRUE),
    avg_population = mean(total_pop, na.rm = TRUE),
    peak_weekly_ci = max(ci_per100k, na.rm = TRUE),
    total_severe = sum(total_severe, na.rm = TRUE)
  ) %>%
  mutate(
    attack_rate_percent = (total_cases / avg_population) * 100,
    severe_rate_percent = (total_severe / total_cases) * 100
  ) %>%
  arrange(desc(attack_rate_percent))

kable(summary_race,
  digits = 2,
  col.names = c("Race/Ethnicity", "Total Cases", "Average Population",
    "Peak Weekly CI (per 100k)", "Total Severe Cases",
    "Attack Rate (%)", "Severe Rate (%)"),
  caption = "Outbreak Summary Statistics by Race/Ethnicity") %>%
kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
  full_width = FALSE)
```

Table 2: Outbreak Summary Statistics by Race/Ethnicity

Race/Ethnicity	Total Cases	Average Population	Peak Weekly CI (per 100k)
American Indian or Alaska Native, Non-Hispanic	22195	158672	1230.21
White, Non-Hispanic	1778774	13848282	1117.81
Black, Non-Hispanic	271836	2211518	1060.00
Hispanic (any race)	1796696	14829946	1052.14
Native Hawaiian or Pacific Islander, Non-Hispanic	16921	153729	962.08
Asian, Non-Hispanic	546770	6295420	760.47
Multiracial (two or more of above races), Non-Hispanic	116391	1611503	617.87

Data Dictionary

Clean Dataset Variables

The final analytical dataset (`fun1`) contains the following key variables:

Variable	Data Type	Description
county	Character	California counties
age_cat	Character	Age category: 0-17, 18-49, 50-64, 65+
sex	Character	Sex: FEMALE, MALE
race_eth	Factor	Race/ethnicity category defined by California Department of Finance
new_infections	Numeric	Number of newly diagnosed individuals during the week
new_severe	Numeric	Newly identified individuals with severe disease requiring hospitalization
cdc_week	Date	Start date of CDC epidemiological week (Sunday)
pop	Numeric	CA Dept of Finance population estimates
ci_per100k	Numeric	Weekly cumulative incidence rate per 100,000 persons
severe_per100k	Numeric	Weekly severe disease rate per 100,000 persons